When Verification Axes Can Support Scientific Claims: Identifiability, Attainability, and Non-Compensatory Promotion

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Abstract

A benchmark score supports a competence claim only when the target is identifiable, compared interfaces can attain its terminal, and the panel is discriminating. We formalize these requirements through fibrewise Bayes risk, a donor-factorization theorem, a terminal-adapter criterion, and data-processing and total-variation bounds. In a protected 420-case battery, a non-compensatory promotion relation made 0/360 false promotions versus 180/360 for the strongest frozen mechanism proxy, while both promoted 60/60 clean positives. A registered abstention contrast was saturated. A distinct exact-axis battery exposed terminal expressiveness rather than general judgement: the governed pipeline selects the undetermined terminal on 30/30 eligible cases with 0/360 false promotions against 0/30 for the same strongest frozen comparator (paired difference 1.0, 95% CI $[1.0, 1.0]$), while the one comparator mechanism able to escalate selects it on 15/30, so the margin against that mechanism is 0.5. On 400 exact contracts, the relation scored 400/400 versus 250/400 for a donor-complete product and 50/400 for a compensatory product; an information-equivalent typed product tied at 400/400. The paired improvement over the donor-complete product was 0.375 with a domain-stratified 95% bootstrap interval of $[0.3275, 0.4225]$. These finite batteries support the factorization and attainability boundary, not naturalistic superiority. A separate source programme has authenticated 76/80 publication–archive–revision bridges but zero eligible natural pairs; an exact 106-class JAR projection does not supply fresh compilation or signed build and mode authority.

Keywords: scientific verification; benchmark identifiability; attainable risk; provenance; authorization; protected evaluation; abstention.

1 Introduction

Language-model research systems increasingly retrieve sources, synthesize claims, attach citations, and run automated verification. This creates a governance problem: a claim may be correct but attributed to the wrong source; a source may support a nearby claim but not the assigned one; a checker may be non-empty yet non-discriminating or share lineage with the answer; and an evaluator may be contaminated or tampered with after the fact. In each case, a scalar confidence score can remain high even though a prerequisite for scientific authority is absent.

We study a stricter object: a *scientific-authority transition.* A proposal can be plausible and well supported but is promoted only when a registered set of content, provenance, checker, and evaluator prerequisites all pass. Unresolved prerequisites do not contribute a low score; they block escalation. This design is motivated by source-aware verification, multi-source attribution, claim-level auditability, evidence influence, iterative fact checking, evaluator-tampering benchmarks, and search-time contamination work (Alvarez et al., 2026; Li et al., 2024; Rasheed et al., 2026; Faizan & Alharthi, 2026; Xie et al., 2024; Atinafu & Cohen, 2026; Wang et al., 2026b; Sadeghi et al., 2026). Recent academic-integrity and agentic-abstention benchmarks also make explicit that declining to complete an unsupported action can itself be the correct behavior rather than a failure to optimize task success (Yang et al., 2026; Liu et al., 2026).

The contribution is deliberately narrower than those component techniques. We ask whether composing these requirements as a *non-compensatory, protected authority transition* materially reduces false scientific-authority promotion while preserving legitimate clean coverage. Throughout, the pipeline built to that rule is called the *governed pipeline*, so that each claim reads as a claim about the rule rather than about a product; Section 5 names the implementation that instantiates it.

The final study was executed after the subject, comparator/ablation mechanisms, evaluator, statistical rules, and split-generation code were frozen. A first 420-case protected campaign on an older subject remains valid evidence for that older revision but does not authorize the repaired one. A separate 39-case live-model arm remains exploratory because post-run audit found label/denominator inconsistencies and a hidden-family execution leak. We therefore generated a fresh post-repair hidden split and executed the publication-authorizing campaign without reusing either prior outcome for tuning.

Our main findings are: (i) the governed pipeline made 0/360 false promotions while the strongest frozen mechanism proxy made 180/360, with the paired H1 interval entirely beyond the predeclared five-point margin; (ii) both promoted 60/60 clean positives, satisfying the H2 non-inferiority guard; (iii) H3 was not supported on that battery, because every system left all eligible cases correctly undetermined on a family that no system could get wrong, while on a repaired battery whose construction clears a fourteen-probe identifiability register the same hypothesis is supported at 1.0 (95% CI $[1.0, 1.0]$) against the strongest comparator's 0/30 and at 0.5 against the 15/30 of the one comparator mechanism that can escalate, which measures the ability to express an undetermined terminal rather than a finer-grained judgement; and (iv) every registered ablation increased false promotion without reducing clean coverage, with soft confidence producing the largest degradation.

2 Related work

Source ownership and attribution. Attribution evaluation across multiple candidate sources is an established measurement problem (Li et al., 2024); source-aware factuality work targets cross-source conflation directly (Alvarez et al., 2026); and scientific claim–evidence benchmarks test whether retrieved evidence supports or contradicts a claim (Javaji et al., 2025). Claim-source retrieval sharpens the same point from the retrieval side, since recovering the source a scientific statement actually came from is a different task from finding a source that would support it (Schreieder et al., 2026). Together these establish claim correctness, source ownership, and semantic support as separate coordinates, which is the premise this paper builds on.

Provenance-sensitive authorization. A current controlled audit makes the authorization distinction especially direct: evidence can be relevant to an agent decision without being authorized to determine that decision, and target-specific tests can hold the task and proposition fixed while varying only source authority (Liao, 2026). The relevance-versus-authorized-evidence distinction is therefore already established. What remains open is the empirical question studied here: whether a protected conjunction of content, ownership, support, checker, influence, evaluator, access, and contamination prerequisites behaves differently from its strongest single component when both meet the same hostile battery.

Auditability, citation fidelity, and influence. Claim-level auditability work argues that research-agent outputs require persistent semantic provenance rather than only fluent text (Rasheed et al., 2026). ProvenAI separates citation fidelity from whether cited evidence influenced the answer path (Faizan & Alharthi, 2026). We adopt that separation: citation presence cannot substitute for evidence-use provenance.

Research-integrity provenance and declared behavior. INSPECT-AI and RIPE-KG represent the provenance of research-integrity assessments over clinical-trial publications, showing a current scientific setting in which the assessment process itself is an auditable object (Markovic et al., 2026). Behavioral Integrity Verification formalizes a complementary declared-versus-actual integrity problem for agent skills (Wu et al., 2026), and certified speculative execution shows the same instinct applied to untrusted agent actions: let the work proceed, but hold its effects behind a check that must clear before they become authoritative (Zhou et al., 2026). These are parent-domain precedents for explicit provenance and contract checking.

Iterative verification and evidence escalation. FIRE iterates retrieval and verification rather than treating a single retrieval pass as final (Xie et al., 2024). DeepSciVerify motivates escalation from an abstract-level judgment to fuller evidence when initial evidence is insufficient (Sadeghi et al., 2026). Stage-wise fact-checking benchmarks make the complementary measurement argument, that an end-to-end verdict hides which stage actually failed (Lin et al., 2026). The first two mechanisms inform frozen comparator proxies here, and the third informs the decision to report per-family and per-coordinate outcomes rather than one aggregate.

Evaluator integrity, benchmark auditing, and contamination. RewardHackingAgents treats evaluator manipulation and held-out leakage as an explicit benchmark dimension (Atinafu & Cohen, 2026). Search-time contamination work shows that browsing systems can retrieve public benchmark answers during inference (Wang et al., 2026b). BenchGuard and Automated Benchmark Auditing show that benchmark specifications, environments, ground truths, and grading logic can themselves be defective (Tu et al., 2026; Wang et al., 2026a). The Holistic Agent Leaderboard further demonstrates the value of standardized harnessing and trace inspection, including direct observation of agents searching for benchmark material rather than solving the intended task (Kapoor et al., 2025). These results motivate the protected evaluator identity, holdout custody, access telemetry, and independent audit used here.

Abstention and scientific integrity. SciIntegrity-Bench constructs academic-integrity dilemmas in which honest acknowledgement of inability is the only correct outcome, while AgentAbstain evaluates paired should-act/should-abstain tasks in executable environments (Yang et al., 2026; Liu et al., 2026). They establish abstention as an evaluation target distinct from generic task success. The object measured here is narrower: an undetermined or blocked terminal is authority-specific, produced by an unresolved or failed evidence, checker, or evaluator prerequisite rather than by a general judgement that the task should not be attempted.

3 Verification-axis identifiability and attainability

A benchmark does not measure a scientific competence merely because it assigns a label and reports an accuracy. The label must be recoverable from the intended scientific information, unavailable from construction nuisances, expressible by the systems being compared, and capable of varying over the comparison panel. These are different requirements. We formalize them before interpreting the protected results.

Let $Y_a \in \mathcal{T}$ be the correct terminal on verification axis a for a random benchmark unit, let D be the representation available to a decision rule, and let finite nonempty $\mathcal{A} \subseteq \mathcal{T}$ be the terminals that the rule is allowed to emit. For a finite benchmark distribution, the smallest attainable exact-terminal error is

$$R^*_a(D, \mathcal{A}) = 1 - \mathbb{E}_D \left[\max_{t \in \mathcal{A}} \Pr(Y_a = t \mid D) \right]. \tag{1}$$

This quantity separates a poor implementation from a task that its information or interface makes impossible.

Proposition 1 (exact axis attainability). There exists a decision rule $g(D) \in \mathcal{A}$ with zero exact-terminal error if and only if every positive-probability fibre of D contains one target terminal and that terminal belongs to $\mathcal{A}$. Equivalently, $Y_a = g(D)$ almost surely for some g with codomain $\mathcal{A}$. This is an almost-sure benchmark statement: it says nothing pointwise about worlds on null fibres.

Proof. On a fibre $D = d$, any rule must emit one terminal. It is correct on the whole fibre exactly when the conditional support of Y_a is the singleton containing that terminal. Summing the best fibrewise choices gives Equation 1, and zero risk gives the stated condition. $\square$

Corollary 1 (donor-product factorization at two authorities). Let D collect the states exposed by a product of provenance, verification, custody, freshness, and generic-authorization donors, and let Y_{promote} be the target scientific-promotion terminal. Pointwise extensional equivalence on a declared world class holds if and only if the target terminal is constant on every donor-state fibre in that class and belongs to the output alphabet. Zero error under a benchmark distribution requires only the corresponding condition on positive-probability fibres; null fibres cannot be certified by a distributional score. If a positive-probability fibre mixes target terminals, every donor-only rule has error at least

$$\sum_{d} \Pr(D = d) \left[1 - \max_{t\in\mathcal{A}} \Pr(Y_{\text{promote}} = t \mid D = d)\right]. \tag{2}$$

The pointwise statement is the set-theoretic factorization argument; the distributional statement is Proposition 1, and the displayed lower bound is attained by a fibrewise Bayes rule on the finite benchmark. Thus a separate module has no inherent advantage once a donor product receives a target-sufficient representation and the same decision relation; conversely, no optimization of a representation that merges differently licensed states can recover the missing distinction. This is a factorization boundary, not a claim that scientific authority must be centralized.

Corollary 2 (terminal-preserving comparator adapters). Let $V \in \mathcal{V}$ be the complete candidate-visible record supplied to an external comparator and let $X \in \mathcal{X}$ be its native output, where $\mathcal{V}$ and $\mathcal{X}$ are standard-Borel spaces. Allow X to be generated by a measurable randomized kernel from V, so that the comparator output adds no world information beyond V. A benchmark adapter is a prospectively declared measurable map $h : \mathcal{X} \to \mathcal{A}$; it is not a post-hoc relabelling rule. Its exact-terminal risk is bounded below by

$$R^*_a(X, \mathcal{A}) = 1 - \mathbb{E}_X\left[\max_{t\in\mathcal{A}} \Pr(Y_a = t \mid X)\right] \;\geq\; R^*_a(V, \mathcal{A}).$$

The first bound is attained by a measurable fixed-order Bayes adapter, and the second inequality is the data-processing consequence of $Y_a - V - X$. Thus a native output is admissible for a three-way exact-axis endpoint with zero benchmark error if and only if its positive-probability output fibres are target-pure and their targets belong to $\mathcal{A}$. For a deterministic native-output map, pointwise admissibility on a declared world class requires the same purity on every native-output fibre, including benchmark-null fibres. In particular, a binary native alphabet cannot attain an endpoint on which all three target terminals occur, and a native Block, "not attributable," parse failure, or free-text insufficiency response cannot be renamed CannotCheck unless the predeclared semantics-preserving map satisfies this fibre condition. When it does not, the mixed-fibre risk is an interface-attainability result and must not be reported as comparator inferiority.

Proposition 2 (claim-level indistinguishability). Let W_0 and W_1 be two scientific worlds in which a proposed competence claim has opposite truth values, and let O be the complete observable record available to an adjudicator. Write $P_i = \mathcal{L}(O \mid W_i)$ for probability measures on one common measurable record space $(\mathcal{O}, \mathcal{F}_O)$, and use the convention $\mathrm{TV}(P_0, P_1) = \sup_{B\in\mathcal{F}_O} |P_0(B) - P_1(B)|$. Under equal prior weight, every measurable, possibly randomized adjudicator has error at least

$$\frac{1 - \mathrm{TV}(\mathcal{L}(O \mid W_0), \mathcal{L}(O \mid W_1))}{2}. \tag{3}$$

The infimum error over adjudicators equals this bound. For two probability measures it is attained: a Hahn decomposition of the finite signed measure $P_1 - P_0$ supplies a maximizing acceptance region. In particular, if the observable records have the same distribution, the competence claim is not identifiable and the error is exactly $1/2$ for every adjudicator.

Proof. For any acceptance region B, the equal-prior error is $\{P_0(B)+P_1(B^c)\}/2$. Taking the infimum over B gives one half of one minus the total-variation distance. The positive set in a Hahn decomposition of $P_1 - P_0$ attains the supremum, while randomized acceptance probabilities cannot improve on it because their risks are mixtures of deterministic risks. □

This second bound concerns interpretation rather than terminal prediction. A score can separate output interfaces while leaving two explanations of that score observationally equivalent—for example, a rule that recognizes an epistemic insufficiency and a rule that is merely forced into the only available non-promoting terminal. A broader competence headline remains unidentified until a discriminator changes the observable law of those worlds.

Nuisance identifiability. Let N contain construction-only quantities such as list length, missingness, template, key order, path name, or label-bearing metadata. The nuisance-only Bayes advantage is

$$I_a(N) = \mathbb{E}_N\left[\max_t \Pr(Y_a = t \mid N)\right] - \max_t \Pr(Y_a = t). \tag{4}$$

A scientific interpretation requires the intended semantic representation to make the terminal attainable while nuisance information contributes no material advantage. A finite shortcut register cannot certify $I_a(N) = 0$ for every possible decoder. It certifies only that the frozen probe class found no such advantage, which is why claim-axis authorization must name the probe class and cannot be promoted to whole-register or naturalistic clearance.

Panel resolution and interface attainability. For a frozen panel $\mathcal{H}$ and score M_a, define its observed resolution as

$$\rho_a(\mathcal{H}) = \max_{h\in\mathcal{H}} M_a(h) - \min_{h\in\mathcal{H}} M_a(h).$$

When $\rho_a = 0$, the experiment supplies no comparative ordering on that axis; it does not establish equality of the underlying competences. When $\rho_a > 0$ but some comparators lack a target terminal in their output alphabet, the contrast establishes interface expressiveness before it establishes the finer judgement of when that terminal is scientifically appropriate. A broad verification-axis claim is therefore licensed only inside the intersection of four conditions: semantic attainability, nuisance resistance, matched terminal attainability, and nonzero panel resolution. Failures fall into distinct research programmes rather than one generic null: repair the construction, enlarge the interface, expose a target-sufficient representation, or improve the decision rule.

Application to the three P4 evidence layers. The historical V2 H3 axis had $\rho_a = 0$: an empty evidence list exposed the gold family and every panel member scored 30/30. That result remains NotSupported. The later V3 construction removes the registered shortcuts and obtains nonzero resolution, but nine of ten comparator mechanisms cannot emit CannotCheck; its exact claim is therefore terminal expressiveness under the frozen gate lattice, not general epistemic judgement.

Stated as an estimand: where a comparator interface cannot emit CannotCheck, the quantity this comparison estimates is *terminal attainability* on the axis, and nothing else. A system that cannot express the correct terminal cannot be observed to reach it, so a gap measured against such an interface is a fact about what the interface can say, not about what the system knows. Reading it as epistemic competence would attribute to judgement what belongs to expressiveness. The P4-X exact-contract successor tests the donor-factorization boundary more directly. Its ideal typed donor product receives the target-sufficient coordinates and predicate and ties exactly at 400/400, while the registered generic-authorization and compensatory products do not. These layers diagnose different limits and are not pooled.

4 Non-escalating authority transition

Let a proposal contain claim c and evidence references $E = \{e_1, \ldots, e_n\}$. Promotion is a conjunction of hard prerequisites rather than a weighted score:

$$G_{\text{content}} : \text{resolved bytes equal the registered evidence digest,} \tag{5}$$
$$G_{\text{owner}} : \text{the assigned source owns the cited content,} \tag{6}$$
$$G_{\text{support}} : \text{the assigned evidence supports } c, \tag{7}$$
$$G_{\text{indep}} : \text{checker lineage is admissibly independent,} \tag{8}$$
$$G_{\text{disc}} : \text{checker rejects the frozen hostile battery,} \tag{9}$$
$$G_{\text{influence}} : \text{cited evidence participated in the answer path,} \tag{10}$$
$$G_{\text{eval}} : \text{evaluator identity/chronology is intact,} \tag{11}$$
$$G_{\text{access}} : \text{protected-label access telemetry is clear,} \tag{12}$$
$$G_{\text{search}} : \text{search-time contamination is clear.} \tag{13}$$

The claim is promoted only when every required gate passes and the evidence supports it. A resolved negative prerequisite blocks promotion; an unresolved evidential prerequisite leaves the claim undetermined. No failed coordinate can be averaged away by confidence elsewhere. Eight ablations remove one coordinate at a time, including a soft-confidence variant that promotes when at least six of nine gates pass.

5 Methods

5.1 Three-valued outcomes

Every authority decision this paper measures is three-valued rather than binary. A claim may be promoted, blocked, or left *undetermined*. An outcome that could not be determined is recorded and counted separately from an outcome determined to be false, and the two are never merged. The distinction is what makes a missing prerequisite legible: an unresolvable digest, an unreachable checker, or an absent provenance record leaves the claim undetermined, and an undetermined claim is not a refuted one. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count.

5.2 The implementation under test

The implementation evaluated here is the ORION authority layer; it is the artifact under test, and the rest of the paper calls it the governed pipeline so that each claim reads as a claim about the mechanism rather than about a product. The results tables and figures label its rows and points by that implementation name, since they are projections of the archived campaign record. Source, protocols, and evidence archives are listed in Section 10.

5.3 Prospective freeze and fresh post-repair split

The base protocol was frozen prospectively, and the post-repair campaign reported here is an additive execution binding on top of it, with a fresh hidden split and a repaired panel; Section 10 names the freeze and the exact subject revision. Before any outcome-bearing run, a public-seed preflight exercised 42 hostile authority tests and the complete 420-case pipeline with ten comparator mechanisms, eight ablations, five deterministic repeats, protected scoring, typed-panel validation, and a separately implemented reproducer. After the preflight and harness merge were signed and green on the default branch, a separate GitHub-hosted job minted a fresh secret-seeded split. The secret seed was never logged; only a commitment was retained.

The final split contains 420 mechanical-gold cases across 13 families: 60 clean positives and 30 cases in each of 12 hostile/insufficient-evidence families. A 150-case post-run public slice contains 30 clean and 120 hostile cases; distinct protected-hostile and protected-holdout slices remain host-only during scoring. Table 1 summarizes custody.

Table 1: Protected V2 battery and custody. Hidden labels remained host-only until scored jobs completed.

Slice	Cases	Clean	Hostile	Candidate labels
Post-run public slice	150	30	120	hidden during execution
Protected hostile	120	0	120	hidden
Protected holdout	150	30	120	hidden
Total	420	60	360	hidden

5.4 Gold-blind execution and custody

The candidate and comparator jobs received exactly two top-level fields per case: an opaque case identifier and the candidate-visible packet. The host mechanically proved that attack family, protected gold, custody class, and expected authority terminal were absent. Candidate and comparator jobs ran in separate GitHub-hosted jobs from the exact subject archive with repository credentials removed. Five deterministic repeats were recorded under file and network syscall tracing. The protected evaluator became eligible to download gold only after both scored lanes completed, and first checked that traces contained neither protected identifiers nor external-IP connections.

The launch workflow also re-queried the signed execution-freeze commit and exact-main CI through the GitHub API, byte-compared the frozen workflow, binding, and harness files to that signed commit, and recomputed host hashes before handing off candidate-visible data. The evaluator and separately implemented reproducer were separate jobs. This is workflow and implementation separation inside one repository, not external evaluator or result-verifier custody.

5.5 Comparator panel, ablations, and statistics

Seven primary comparator mechanisms mirror the repaired panel: ProvenanceGuard-style source routing, AttributionBench-style multi-source attribution, FIRE-style iterative retrieve/verify, CLAIM-BENCH-style claim evidence, ProvenAI-style citation fidelity/influence, RewardHackingAgents-style contamination/tamper checks, and DeepSciVerify-style evidence escalation. Three auxiliary controls test citation presence, pooled semantic support, and auditability/provenance.

Comparator names therefore use “-style” or describe a mechanism. They are protocol-matched reimplementations under one candidate-visible packet and resource accounting, not executions of the external authors’ software. Every empirical claim below is about the frozen mechanisms in this benchmark, not about published systems in their native environments.

The primary outcome is false authority-promotion rate over gold terminals that do not permit promotion. Clean coverage is promotion rate over the 60 clean positives. H1 requires an absolute false-promotion improvement beyond 0.05 against the strongest frozen comparator; H2 requires clean coverage non-inferior within 0.05; H3 tests superior selection of the undetermined terminal where that is the correct answer. Rate intervals use Wilson 95% intervals; paired effects use a fixed-seed percentile bootstrap. No false promotion, false block, abstention, candidate-caused failure, or clean false negative is removed from its eligible denominator.

6 Results

The typed authority panel passed with no blockers, and a separately written repository implementation reproduced the headline counts. This is a local reproduction, not external verification authority.

6.1 False promotion and clean coverage

The governed pipeline produced 0/360 false promotions. The strongest frozen comparator mechanism, ProvenAI-style citation-fidelity/influence, produced 180/360 (50.0%). The paired difference was -0.50 with 95% CI $[-0.553, -0.447]$, so H1 passed the predeclared -0.05 practical margin. Figure 1 reports the full mechanism panel.

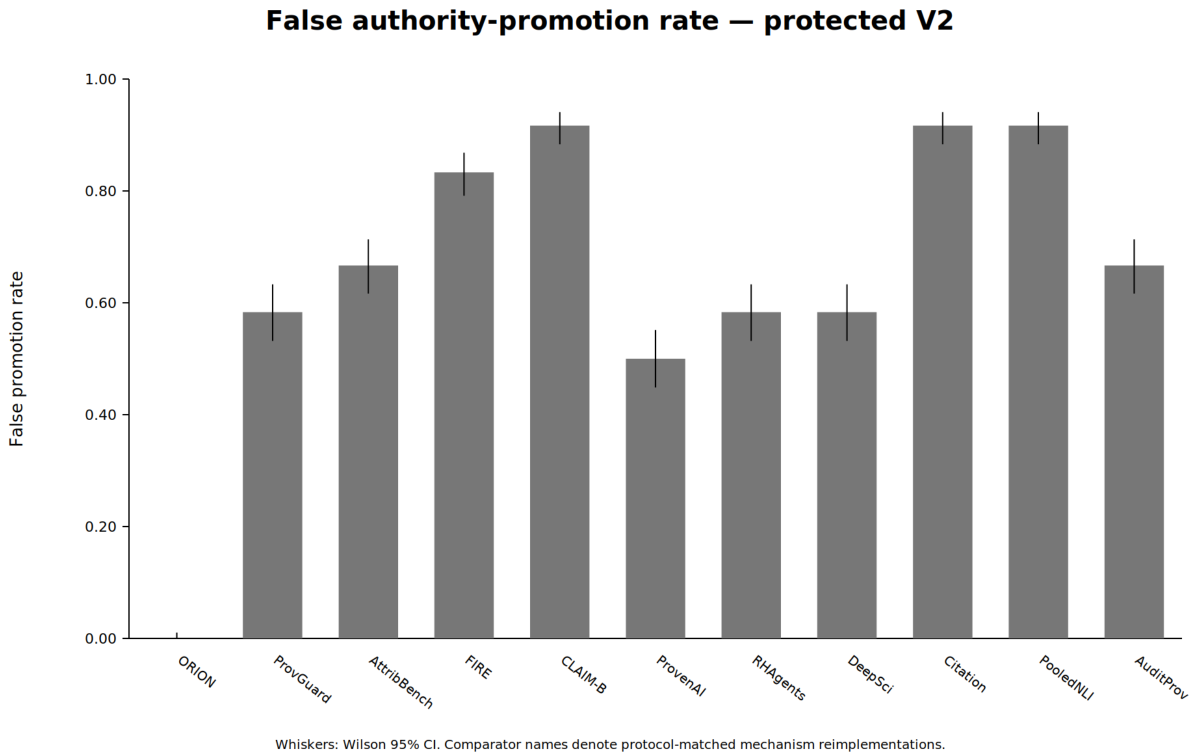

Figure 1: False authority-promotion rate in the protected campaign. Whiskers are Wilson 95% intervals. Comparator labels denote protocol-matched mechanism reimplementations.

Table 2: Fail-closed outcomes and false-negative cost in protected V2. The only gold family whose correct terminal is 'undetermined' contains 30 insufficient-evidence cases.

System	Correctly undetermined	False promotions	Clean FN
ORION	30/30	0/360	0/60
ProvenAI-style	30/30	180/360	0/60

Both the governed pipeline and the strongest comparator promoted 60/60 clean positives, with zero clean false negatives. The clean-coverage difference and paired 95% interval were both exactly zero, so H2 passed the -0.05 non-inferiority margin. Figure 2 shows that every primary mechanism sits at 1.0 clean coverage; the result therefore does not manufacture a frontier by visually jittering equal values.

On this battery H3 was not supported, and the cause is in the battery rather than in the systems. Every one of the eleven panel members, the governed pipeline included, correctly left all 30 insufficient-evidence cases undetermined, so the difference is zero with 95% CI $[0, 0]$, and the ablations do not instrument the metric at all.

The construction explains why. In this battery the insufficient-evidence family was generated with an *empty* evidence list, while every other family carried exactly one evidence object. The rule "leave the claim undetermined if and only if the evidence list is empty" therefore classifies all 420 cases correctly. Detecting an empty list is not an act of judgement, so the metric was saturated by construction and the comparison had no resolving power: a difference could not have been observed had one existed. We report this as an inability to discriminate rather than as evidence that no difference exists, and we do not treat the null as a finding about abstention. Table 2 reports abstention and false-negative cost on that basis.

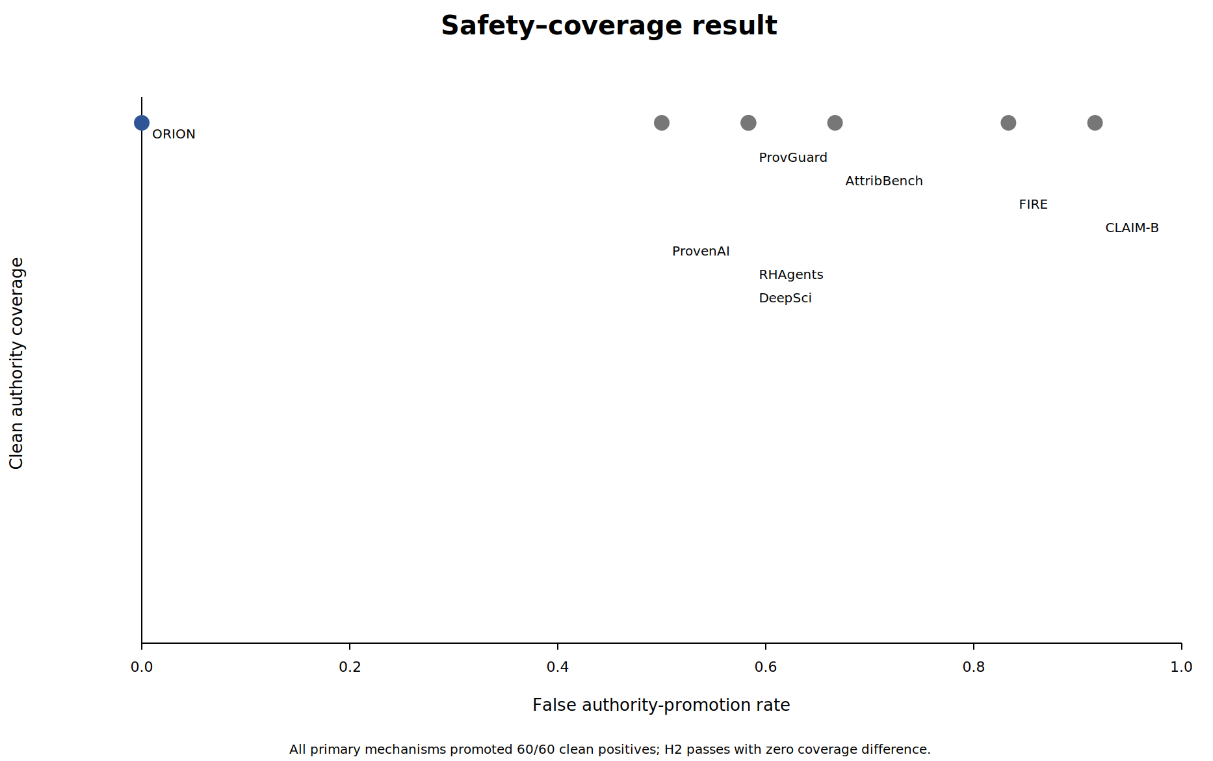

Figure 2: Safety–coverage result. Every primary mechanism promoted 60/60 clean controls, while hostile false-promotion rates differ substantially.

6.2 Repairing the construction, and one run of the frozen panel on it

The first repair changed what must be reasoned about rather than any gold label: cases present evidence that is read but does not establish the claim, in two subtypes, one where the pool contains no supporting record at all and one where a supporting record is present but carries no verifiable provenance. That removed the empty list and did not remove the shortcut. The repaired bodies took five distinct lengths, two of which occur only on undetermined cases, so a rule over evidence-body length recovers the terminal on 20/20 protected cases with no false positives among the other 250 — and the family named itself in English besides. Measured after the fact with the register described below, five of fourteen judgement-free probes recover the undetermined label at informedness 1.00 on that construction, and thirteen of the fourteen do so on the published one. Repairing a named shortcut is not the same as establishing that there is none, so we do not report a panel from it.

The second repair was written against the property rather than against a named cue, and its protocol, seeds and interpretation were frozen in advance of the repair and of any panel outcome. Every case in the battery is shape-identical: one evidence-body template for all 840 records, with the same length, word count, punctuation and character-class profile; fixed container sizes throughout; no absent field anywhere; and support carried by a token that is present on every record, so that only token matching separates support from non-support. The undetermined family is split into two shape-identical subtypes and is a strict subset of a set that also contains 30 blocking cases, so neither "nothing in the pool supports" nor "the assigned record does not support" classifies it. Families differ only in the values of the fields an authority gate is defined over. The identifiability register — fourteen judgement-free probes over counts, container lengths, key shapes, string lengths, character-class and template fingerprints, identifier shapes and scalar values, fitted on the fitting split and scored on the protected one — was run and recorded before the panel was invoked. On the repaired construction it reports informedness 0.0 for every probe on the undetermined axis, on all

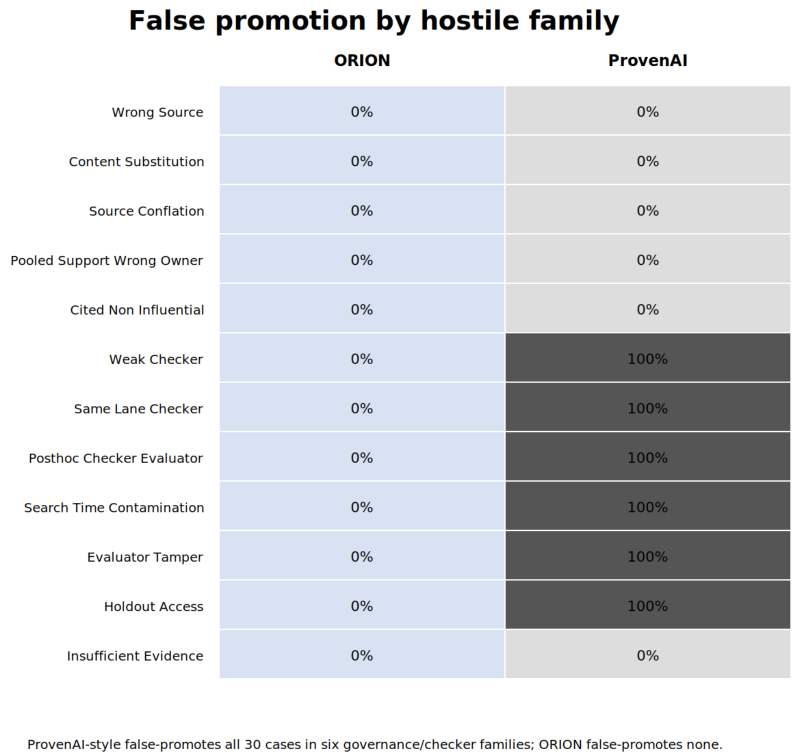

Figure 3: False-promotion rate by hostile family for the governed pipeline and the strongest frozen mechanism proxy.

thirteen registered seeds, against a declared ceiling of 0.0; not one probe ever predicts the undetermined terminal for any protected case.

Run once on that battery, the same frozen panel and statistics report H3 supported. The governed pipeline selects the undetermined terminal on 30/30 eligible cases with 0/360 false promotions; the strongest frozen comparator, selected by H1 exactly as before, selects it on 0/30; the paired difference is 1.0 with 95% CI $[1.0, 1.0]$ against a predeclared threshold of a positive lower bound. The one comparator mechanism with an escalation path, DeepSciVerify-style abstract-to-full escalation, selects it on 15/30, so the margin against that mechanism is 0.5 rather than 1.0; both numbers are needed to read the result. What the difference measures was fixed before the panel ran and is narrower than the headline: terminal expressiveness under a non-compensatory gate lattice — the ability to report an inability — and not a finer-grained scientific judgement. Nine of the ten comparator mechanisms score zero because, once the empty-evidence case is gone, they cannot express an undetermined terminal at all, and a panel of eleven systems in which ten are two-valued cannot separate that reading from a claim about knowing when one cannot check. Clean coverage is 1.0 for all eleven systems on the repaired battery as well, so H2 remains a saturated axis there and is still not a comparative finding. The V2 numbers above are unchanged and are not restated by this run: it is a different battery, and its result does not license reinterpreting the null the V2 construction produced.

6.3 Attack families and ablations

The governed pipeline made no false promotion in any hostile family. The strongest comparator false-promoted all 30 cases in each of six governance/checker families: weak checker, same-lane checker, post-hoc checker/evaluator, search-time contamination, evaluator tamper, and holdout access. It was correct on content substitution, source conflation, wrong-source, pooled-owner, cited-non-influential, and insufficient-evidence families. Figure 3 shows the contrast.

Table 3: Protected V2 results. Comparator names denote protocol-matched mechanism reimplementations, not original external software. FP is over 360 hostile opportunities; clean is over 60 clean positives.

System / variant	FP	Clean	Clean FN	Latency (ms)	Resource
ORION	0.000	1.000	0	2.9736	9.0
ProvenanceGuard-style	0.583	1.000	0	0.0037	5.0
AttributionBench-style	0.667	1.000	0	0.0058	3.0
FIRE-style	0.833	1.000	0	0.0069	6.0
CLAIM-BENCH-style	0.917	1.000	0	0.0049	4.0
ProvenAI-style	0.500	1.000	0	0.0034	5.0
RewardHackingAgents-style	0.583	1.000	0	0.0072	5.0
DeepSciVerify-style	0.583	1.000	0	0.0074	6.5
Citation presence	0.917	1.000	0	0.0009	1.0
Pooled support	0.917	1.000	0	0.0049	2.0
Auditability/provenance	0.667	1.000	0	0.0033	5.0
ORION - exact content	0.083	1.000	0	2.7131	9.0
ORION - source/provenance	0.167	1.000	0	2.7077	9.0
ORION - checker lineage	0.083	1.000	0	2.7078	9.0
ORION - hostile checker	0.083	1.000	0	2.7043	9.0
ORION - influence	0.083	1.000	0	2.7045	9.0
ORION - evaluator protection	0.250	1.000	0	2.7116	9.0
ORION soft confidence	0.917	1.000	0	2.4570	9.0
ORION - search block	0.083	1.000	0	2.7087	9.0

Every registered ablation preserved 60/60 clean coverage while increasing false promotion. Removing exact content binding, checker-lineage independence, hostile-checker discrimination, behavioral influence, or the search-contamination block produced 30/360 false promotions (8.3%) each. Collapsing source/provenance identity produced 60/360 (16.7%); removing evaluator protection/telemetry produced 90/360 (25.0%); and replacing the fail-closed terminal with the predeclared six-of-nine soft-confidence threshold produced 330/360 (91.7%). Table 3 contains the complete panel.

6.4 Attribution, support, cost, and telemetry

Several mechanisms achieve perfect source-attribution accuracy while still false-promoting governance attacks; conversely, pooled semantic-support mechanisms can score well on support while attributing the source incorrectly. Figure 4 separates those coordinates.

Deterministic gate evaluation is slower than the simple mechanism proxies in this synthetic harness: mean latency is about 2.97 ms per case for the governed pipeline versus single-digit microseconds for the primary proxies (Figure 5). This is not an end-to-end serving benchmark; model and network latency are intentionally absent.

Scored-process telemetry recorded zero protected-identifier hits and zero external-IP connections for both candidate and comparator jobs. Raw traces remain in protected custody. The separate repository reproducer obtained the same 0/360 versus 180/360 false-promotion counts and 60/60 versus 60/60 clean-promotion counts; this code-path separation does not supply external result-verifier custody.

6.5 Exclusion of exploratory evidence

We exclude the repository's earlier 39-case live-model arm from publication authorization. Post-run audit found internally inconsistent live labels/expected terminals, incorrect opportunity denominators in its summary path, and direct use of a candidate-hidden family label in the live execution path for the governed pipeline and its ablations. Those outputs remain useful diagnostics but are not combined with, substituted for, or used to tune the protected campaign.

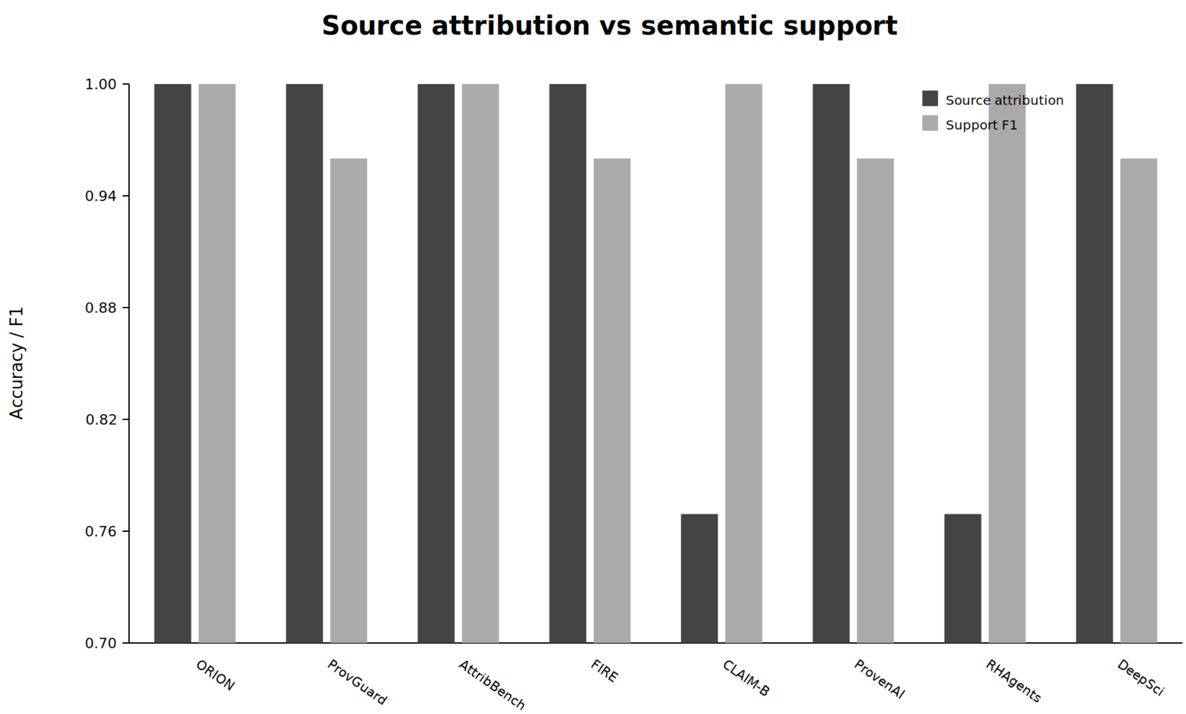

Figure 4: Source-attribution accuracy and support/contradiction F1 for the primary mechanism panel. High values on these coordinates do not guarantee a safe authority terminal.

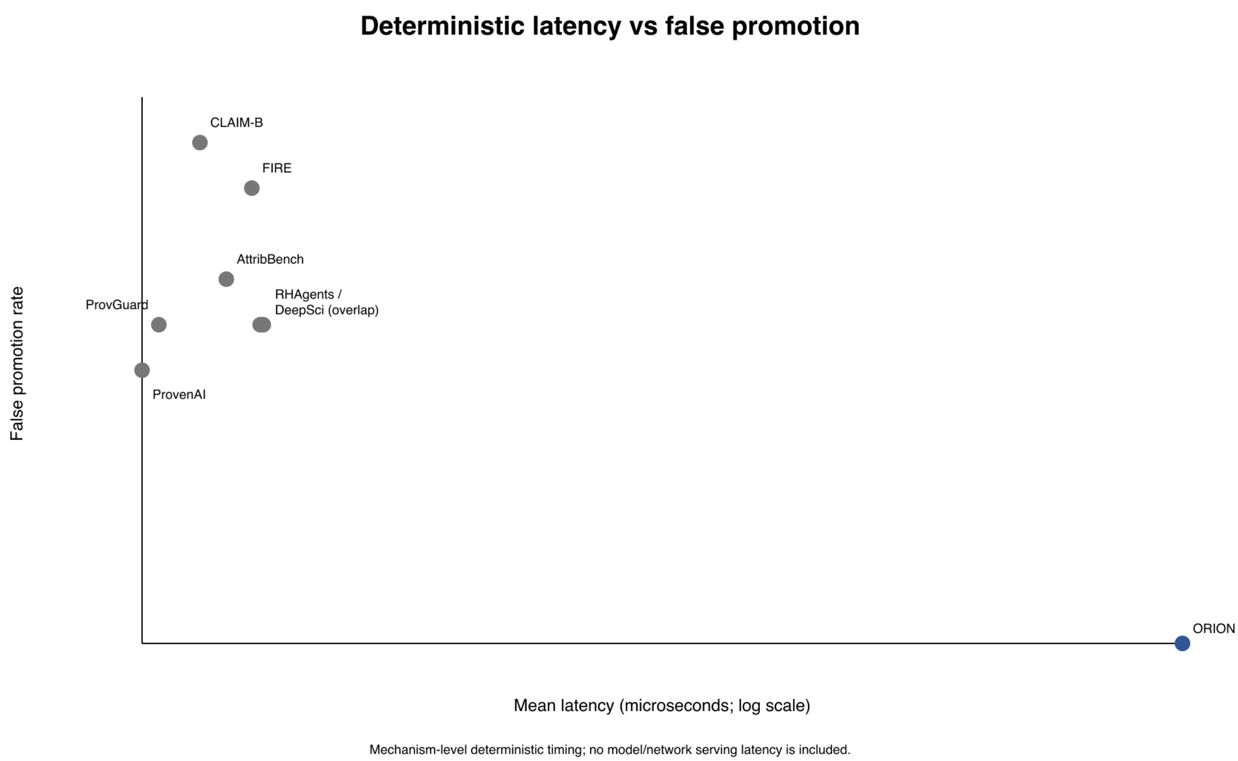

Figure 5: Mean deterministic evaluation latency versus false authority-promotion rate; the latency axis is logarithmic.

7 Post-saturation successor: donor-complete scientific-promotion authority

The protected V2 result shows that a non-compensatory content/provenance/checker/evaluator pipeline can eliminate false scientific-authority promotion relative to strong frozen mechanism proxies while preserving every clean positive. P4-X raises the comparison standard. Ordinary provenance, claim verification, artifact/version binding, evaluator custody, epoch handling, and generic authorization are granted to a donor-complete product, leaving only the final *scientific-promotion relation* as the discriminator. The V2 and successor campaigns remain separate prospectively frozen experiments and are not pooled.

7.1 Donor-engulfment architecture

P4 retains source-aware factuality, provenance, claim/evidence verification, citation fidelity, evaluator auditing, contamination defense, abstention, and generic authorization as donor-owned local components (Alvarez et al., 2026; Faizan & Alharthi, 2026; Liao, 2026; Tu et al., 2026; Zhou et al., 2026). P4-X tests the higher-level authority relation that determines whether those successful local verdicts are sufficient to change the scientific standing of a target claim.

The P4-X relation is non-compensatory. A target claim is promoted only when every registered hard scientific-promotion obligation is discharged. In the exact successor battery, the discriminating coordinates concern target claim scope, independence of supporting evidence, and unresolved scientific-promotion authority after the ordinary provenance, verification, custody, and authorization checks available to the donor product have already succeeded.

7.2 Exact successor design

The frozen successor contains 400 protected cases spanning five heterogeneous scientific artifact domains, eight hostile/control archetypes, and ten protected variants per domain–archetype cell. All systems receive the same candidate-visible information.

The principal arms are:

- **P4-X**: the non-compensatory scientific-promotion relation;
- **B1**: a strong donor-complete generic-authorization product with ordinary provenance, verification, artifact/version, custody, and epoch information but without the complete target-bound scientific-promotion relation;
- **B2**: a compensatory all-signal product that permits successful coordinates to offset failed or unresolved ones;
- **B3**: an ideal typed donor product given every P4-X scientific coordinate and the same promotion predicate, testing portability of the semantics.

7.3 Protected result: perfect exact promotion decisions

P4-X selects the correct scientific-promotion terminal on **400/400 protected cases**. B1 is correct on 250/400, B2 on 50/400, and B3 on 400/400 with zero decision mismatches to P4-X. The paired P4-X minus B1 effect is +0.375 with a domain-stratified bootstrap 95% CI $[0.3275, 0.4225]$; paired McNemar discordance is 150–0. P4-X makes **zero false promotions**, retains clean promotion at 1.0, and its advantage over B1 is positive in every exact domain. An independent implementation that does not import the P4-X execution scorer reproduces the canonical protected-row digest and all arm counts.

The result isolates a specific architectural gain above strong donor machinery. Ordinary provenance, verification, custody, and generic permission can all be valid while claim scope, evidence dependence, or an unresolved scientific obligation still prevents promotion. Making the target-bound promotion relation explicit closes all 150 such B1 errors without sacrificing any clean promotion.

7.4 Portability result

B3 also reaches 400/400 and is decision-identical to P4-X. This is a positive systems result: the scientific-promotion semantics are sufficient and portable. They can be implemented as an ORION layer or as a correctly enriched donor product without changing any registered decision. The P4-X advantage over B1 therefore identifies the missing scientific coupling in a realistic donor-complete interface rather than depending on centralized expressivity.

7.5 Measurement-resolution audit

The successor does not retroactively reinterpret the historical V2 secondary metrics. V2 H2 is a clean-coverage guard in a battery where all primary systems achieve perfect clean coverage, confirming that the false-promotion result is not blanket refusal. V2 H3 is non-discriminating because the original insufficient-evidence construction is structurally saturated. The post-run audit identifies that shortcut and prevents a trivial equality from becoming a scientific equivalence claim. This instrument diagnosis strengthens the authority discipline of the paper while leaving the protected V2 and P4-X headline results unchanged.

7.6 Strongest supported statement

After donor-complete provenance, verification, custody, artifact binding, freshness, and generic authorization are granted, *scientific claim promotion remains a distinct non-compensatory authority relation.* On the registered heterogeneous exact contracts, making that relation explicit raises exact promotion accuracy from 0.625 to 1.000 against the strong donor-complete product while preserving zero false promotions and perfect clean coverage. Live verifier/provider transfer is the next empirical stress test of this demonstrated authority relation.

8 Threat model, limitations, and interpretation

The theory and the evidence have different scopes. Proposition 1 and its factorization corollary hold for any finite exact-terminal decision problem, but they identify what information and interface make a decision attainable; they do not show that a deployed verifier possesses the target-sufficient representation. The empirical layers establish only the representations, relations, and case distributions that were frozen for those layers.

First, the V2/V3 campaigns use 420 synthetic mechanical-gold cases; they do not establish performance on naturally occurring scientific disagreements. The written human-adjudication rubric was never triggered because all final cases were mechanically decidable, so no claim is made about adjudicator agreement. P4-X adds five heterogeneous artifact domains and eight archetypes but remains a 400-case exact-contract study, not a population sample of scientific disputes. Second, scoring is deterministic and uses no LLM provider. It isolates authority-transition semantics but does not measure the error distribution of a full stochastic research agent.

Third, comparator mechanisms are matched reimplementations. They instantiate published ideas under one packet and budget, but the result must not be read as a leaderboard against the original ProvenAI, ProvenanceGuard, DeepSciVerify, FIRE, AttributionBench, CLAIM-BENCH, RewardHackingAgents, or related implementations. Fourth, the clean controls are intentionally unambiguous; 60/60 coverage shows that the governed pipeline did not win by blanket refusal but does not estimate recall on difficult benign cases. Fifth, the hostile taxonomy is finite. Adaptive attacks, compromised host infrastructure, side channels outside retained syscall/network telemetry, and novel evaluator attacks remain outside this campaign.

Sixth, and most consequential for how the three hypotheses should be read, two of them are measured on axes that this battery saturates, and for H3 the mechanism is identified rather than suspected. Clean coverage is 1.0 for all eleven panel systems and for all eight ablations, and the correct-undetermined rate is 1.0 for all eleven systems and is not instrumented in the ablations at all. Only false authority promotion varies: it takes six distinct values across the panel and four across the ablations. The whole empirical separation reported here rests on that one axis. H1 is therefore a measurement; H2's non-inferiority pass and H3's null are statements about a battery in which neither quantity was ever observed to move. Testing them would require clean positives that a weaker governance mechanism actually declines, and insufficient-evidence cases that a weaker mechanism actually over-promotes or over-abstains on — cases the current gold construction does not contain. On this battery, therefore, H2 and H3 should be read as design limits rather than as comparative findings.

For H3 such cases now exist, and Section 6 reports them: on the repaired battery the correct-undetermined rate takes three values across the same eleven systems, so the axis moves and the hypothesis is decided by the systems rather than by the construction. Three things that result does not do. It does not retrospectively license the null above — the repaired battery is a different battery, and the numbers reported for the frozen campaign stand as the record of what that construction produced. It does not widen what H3 claims: nine of the ten comparator mechanisms score zero because they cannot express an undetermined terminal at all, so the quantity measured is terminal expressiveness under a non-compensatory gate lattice, and the margin is 1.0 against the strongest comparator but 0.5 against the one mechanism that can escalate. And it does nothing for H2, whose clean coverage is 1.0 for every system on both batteries; that axis remains a design limit with no case yet built that could move it.

The V3 nuisance audit is also claim-axis specific. Its fourteen frozen probes and thirteen seeds clear the exact CannotCheck axis at informedness 0.0, while the adjudication receipt retains four off-axis residuals and explicitly withholds whole-register clearance. More importantly, nine of ten comparator mechanisms cannot emit CannotCheck. In the terminology of Section 3, V3 has nonzero observed resolution but unmatched terminal attainability. It therefore repairs the V2 construction failure without yet supporting a general scientific-judgement claim.

Seventh, the mechanisms composed here are not claimed as novel, and the boundary is worth stating once and plainly. Provenance tracking, attribution evaluation, citation checking, iterative verification, evidence escalation, contamination detection, evaluator-tamper detection, abstention, benchmark auditing, claim-level auditability, provenance ontologies, and generic behavioral-integrity checking are all prior work, cited in Section 2. The relevance-versus-authorized-evidence distinction is likewise established there. What is measured here is the behavior of the non-compensatory conjunction of those prerequisites under a protected, gold-blind battery, and nothing beyond it.

P4-X does not remove that boundary. It grants the strongest comparison product ordinary provenance, verification, exact artifact/version binding, evaluator custody, epoch handling, and generic authorization before testing the target scientific-promotion relation. Its 400/400 versus 250/400 result shows that the registered generic relation is insufficient on the exact successor contracts; the 400/400 ideal-product tie simultaneously shows that an information-equivalent donor product is sufficient. Neither count licenses a centralization advantage, universal necessity of the exact gate list, or superiority over deployed donor systems.

Recent benchmark-auditing work is an additional caution: expert-built tasks and evaluation harnesses can contain specification, environment, ground-truth, or grading defects that materially alter measured rankings (Tu et al., 2026; Wang et al., 2026a; Kapoor et al., 2025). Deterministic generation, typed-panel validation, exact split/harness binding, protected custody, post-run public slicing, and independent reproduction mitigate this risk, but these controls do not prove that the 13-family taxonomy is complete or that every benchmark-design assumption is correct.

The reported intervals are computed at the case level, and the cases are not independent: they cluster by attack family. A cluster-respecting reanalysis of the protected campaign, resampling families rather than cases over 20,000 replicates, widens the normal half-width from 0.0528 to 0.2829, a ratio of 5.48, and gives a family-bootstrap interval of $[-0.75, -0.25]$ against a published case-level interval of $[-0.553, -0.447]$. The direction and sign of the effect survive that change of inference unit, and an exact sign test over the twelve independent clusters is two-sided $p = 0.031$ with six discordant clusters, all six favouring the governed pipeline and none favouring the comparator. What does not survive is the apparent precision: any case-level interval in this paper should be read as roughly five times narrower than the family-level evidence supports, and the effect is established in direction rather than in magnitude.

The registered naturalistic successor, identified as `P4.H3.NATURALISTIC.V1`, is the upward test rather than a claimed result. It specifies a target of 768 independent source–case-family clusters in 32 strata (four domains by eight unidentifiability mechanisms), with each unidentifiable unit paired to an identifiable same-source control, a worst-domain gate, independent evaluator/raw-case custody and source-disjoint replication. The clusters do not yet exist. The first outcome-blind preflight fixed twelve GitHub software roots. A successor now fixes 16 candidate roots across two waves, with four metadata providers, four artifact modalities, four nominal domains and two roots per domain in each wave. Its 13/13 conformance checks and 16/16 live identity checks pass, but it opened no case text and created no exact-claim-axis pair, mechanism label, blinding or replication panel. Eight roots lack exact historical Crossref bytes, while content-class rights and case eligibility remain unresolved for all sixteen.

A deeper single-provider source frame now binds 1,536 exact-version arXiv records, 384 in each domain, only where the official record reports CC BY 4.0. That closes candidate identity, attributed article-text reuse and byte fixity of the metadata pool, not a case panel. The identity-complete V2 amendment names all eight natural information-state mechanisms and allocates 24 primary, eight source-disjoint replication and sixteen reserve candidates per domain–mechanism cell. Its outcome-free metadata census exposes a source-universe negative: only abstract–full-text and earlier–later-version signals reach the nominal 48 records in every domain. The lowest-domain signal counts for protocol–result, article–correction, article–data, article–code, conference–paper and article–supplement pairs are respectively 2, 5, 38, 6, 16 and 2. Since these signals are upper bounds on linked, rights-cleared eligible pairs, the arXiv frame cannot fill the declared 32-cell panel. A source-expansion successor is frozen; spare cases may not be reassigned to convenient mechanisms.

The first Zenodo source-expansion identity is itself a retained transport failure: its frozen page size of 200 was invalid for unauthenticated requests, so it produced no census. The distinct V2 repair passed all response-schema and pagination checks, but its publication-typed signal counts remained below 48 in all eight frozen cells: 29–44 for the four article–data queries and 6–10 for article–software. The latter shortfall persists despite abundant software relations because most point to non-publication targets. These 98 unique public records are discovery candidates only. Their record licences and public file links do not bind the licence, identity or scientific relation of the linked publication, and no metadata rule can replace external natural-pair adjudication. DataCite, repository-release/Software-Heritage and PMC OA routes therefore remain disjoint next discriminators rather than post-hoc relaxations of the Zenodo gate.

The first DataCite M5 identity also remains adverse: its sparse response omitted the client relationship required by the frozen provider-disjointness rule, so all 4,000 rows failed closed before scientific screening. A distinct V2 fixes that mechanics error and excludes every V1-disclosed DOI. It yields 3,032 provider-disjoint rows and 2,719 exact CC BY 4.0/CC0 declarations, but no HTTPS `contentUrl`; hence zero rows reach relation adjudication and the M5 deficits remain 11, 4, 19 and 18 across the four domains. This is not evidence that DataCite lacks publication relations: the relation count was not reached after the earlier availability gate failed. DataCite metadata CC0 does not license repository files, a depositor rights declaration does not prove authority over those files, and URL visibility would not itself establish permission. A future successor must bind a repository-specific public-file manifest or exact record API without dropping the file-identity or rights gates. Until then linked-publication rights, natural-pair identity and eligibility remain CannotCheck or unadjudicated.

The later Dryad/Figshare/Harvard Dataverse/DataCite expansion does not yet turn those deficits into a population conclusion. Figshare and Dataverse transport captured only 659/2,463 and 512/1,540 frozen full records; their missing identities cannot be replaced after the outcome-free freeze. Dryad download authorization and exact version/file lineage remain undetermined, and DataCite discovery metadata contributed no independent repository-bound unit. The observed strict counts are lower bounds, so both a claim that the cells fail and a claim that they pass would exceed the evidence. Completion must resume the frozen missing identities or open a separately frozen provider family; it may not add convenient records to the current census.

The bounded JOSS/GitHub V4 route closes transport only for its frozen 200-publication frame. Its no-extension V5 successor recovers 9/9 exact V3 M6 identities for DOI-level deduplication and binds 39/80 exact publication–archive-version–tag–immutable-commit relations. The distinct same-identity V6 repair resolves 17 of V5's 41 failures by content-addressed archive-manifest/commit or qualified-SWHID/Git-tree equality, without adding or replacing a DOI unit. A targeted V7 source-native successor then repairs 14 of those 24 without extending the publication frame, raising the exact bridge to 70/80. Ten remain unresolved: three archive-to-commit authenticated-identity failures, two exact-archive-rights failures, two exact archive-version DOI failures and three exact tag-to-immutable-commit failures under the primary cause partition. Public author and repository-owner metadata still do not establish author-lineage independence, same-claim preservation, a single information-coordinate intervention or scientific resolvability. Thus author-lineage adjudications and natural pairs remain 0.

The V6 exact deduplicated frame was Earth 6/48, life 4/48, scientific software 51/48 and physical 4/48, with Figshare replication sides 2/8, 0/8, 6/8 and 1/8. V7 repairs identity links only; it does not create source-disjoint replication, author-lineage evidence or eligible natural pairs. Resolving the remaining same-200 bridge failures cannot by itself close the three non-software cells or the software replication side within this frame. Separately frozen non-GitHub domain-provider routes and external author-lineage/natural-pair adjudication are required; packages, releases, files, versions, tags and commits may not be multiplied into independent scientific units.

V8 removes three of the ten residual identities with provider-native archive, version, provenance and licence evidence, reaching 73/80 exact bridges. The remaining seven require authoritative edges not recoverable by repeating the same archive download: three archive-to-commit identities, three tag-to-full- commit/archive identities and one inconsistent version/root-to-commit chain. This sharper queue is still identity preflight, not a naturalistic panel.

V9 exact-content comparison repairs two of those seven, preserving a historical 75/80 stage. V10/V10B repairs one further content-identity edge only after the original V10 target-specific discriminator fails and a distinct outcome-informed successor is frozen before new downloads. The resulting 76/80 bridge leaves four exact failures: two require provider correction of the frozen archive relation and two require provider-native distribution-to-commit build provenance. Exact-content identity does not establish version ancestry, source-disjoint replication, author-lineage independence, natural-pair eligibility or external custody.

V11 confirms that the same four edges cannot be closed by repeating unchanged provider requests. V12 then proves that the 106 untracked Java classes in index 91 are exactly available inside a tracked JAR, but fresh source compilation remains CannotCheck. This byte equality is insufficient because no provider-native signed build statement binds the JAR projection to the released archive, and the two archive executable modes contradict the validly signed 100644 revision tree. The positive projection therefore narrows the provenance residual without changing the 76/80 bridge, natural-pair eligibility, replication, custody or outcome authority.

The V2 amendment now enumerates four comparator roles—calibrated three-way NLI, scientific evidence escalation, provenance-aware verification, and an information-equivalent typed boundary product—rather than leaving four anonymous slots. This is design completion, not execution binding: exact model or code revisions, licences, entrypoints, budgets, calibration and separately implemented records remain missing. No audited neighbour natively separates ResolvedTrue, ResolvedFalse and CannotCheck: binary Block, "not attributable," parse failure or free-text insufficiency cannot be relabelled into the third terminal without satisfying the adapter fibre condition. The current terminals are `P4_NATURALISTIC_V2_IDENTITY_COMPLETE__FEASIBILITY_AND_EXTERNAL_PANEL_CANNOT_CHECK` and `P4_EXTERNAL_TERMINAL_CANNOT_CHECK`. No naturalistic, cross-domain, external-comparator, or source-disjoint result is inferred from the protocol, metadata snapshot or local entrypoint probes.

The appropriate current interpretation is a verification-axis envelope. V2 measures false-promotion separation and records a saturated negative axis; V3 shows that a shortcut-resistant exact axis can still be interface-limited; and P4-X shows both failure of a registered generic-authorization relation and exact equivalence when a donor product receives target-sufficient state and the same predicate. Together they support a fail-closed scientific-promotion interface and the factorization theory that explains its limits. They do not prove universal scientific truth, naturalistic scientific judgement, general abstention ability, or evaluator security.

Repository-level integrity is not scientific authority. The hashes, frozen digests, deterministic replays and receipt chains this paper relies on establish that a stated computation was the computation performed and can be performed again. They establish nothing about whether its conclusion is correct, whether the benchmark measures what it claims, or whether an independent party has reviewed either. A perfectly replayed run of a flawed protocol replays the flaw exactly. Where this paper's claims require external scientific adjudication, that adjudication has not occurred and the integrity record does not substitute for it.

9 Broader impact and governance

False scientific-authority promotion can amplify incorrect or misattributed claims, particularly when a system presents verification artifacts that appear independent but are not. A fail-closed layer may reduce this risk, but excessive blocking can delay legitimate scientific communication or concentrate power in whoever controls evidence and evaluator registries. Work on academic integrity and agentic abstention reinforces that refusal can be the correct behavior under genuine insufficiency, while also showing that productive abstention remains difficult and should not be conflated with generic non-response (Yang et al., 2026; Liu et al., 2026). We therefore report abstention cost, clean coverage, custody identity, and artifact transparency rather than optimizing only a security metric.

Protected evaluation creates another trade-off: some labels must remain hidden during scoring, yet publication requires enough evidence for audit. We release safe aggregates and a post-run public slice while retaining protected holdout labels/traces in restricted custody. Exact subject/split/harness hashes and a separate-implementation reproduction receipt are published so the result is not represented by an unverifiable scalar score. This receipt supplies repository code-path separation, not external result-verifier custody.

10 Data and code availability

Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative. Source, protocols, and evidence are version-controlled; the repository is withheld during double-blind review and will be named in the camera-ready version, where a persistent DOI will also be assigned. All paths below are relative to the paper's directory in that repository.

10.1 Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P4.protected-authority.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number reported here can be traced to a decision made in advance of seeing it.

The machine-readable freeze is `protocol/PROTOCOL_V1.json`, the case requirements are `protocol/ATTACK_CASE_SCHEMA_V1.json`, and the statistical rules, metric definitions, and threat model are `protocol/STATISTICAL_ANALYSIS_PLAN_V1.md`, `protocol/METRICS_REGISTRY_V1.json`, and `protocol/THREAT_MODEL_V1.md`. The custody policy and the execution-freeze checklist for the post-repair campaign are `protocol/CUSTODY_POLICY_V1.md` and `protocol/EXECUTION_FREEZE_CHECKLIST_V2.md`. The prospective execution binding, including the exact subject revision and the frozen harness identity, is `protocol/PROTECTED_RUN_BINDINGS_V2.json`. The comparator and ablation disclosure is `host/BASELINE_CONFIGS_V2.json`, and the frozen case generator and verdict-leak check are `protocol/generate_attack_manifest.py` and `protocol/check_verdict_leak_v1.py`.

10.2 Executed campaign and result artifacts

The publication-authorizing campaign is GitHub Actions run `31976589735`. The subject under test is commit `f6e51b5c8f905382b8e2f5568d9035fc14241aa1`; the split SHA-256 is `3fe91b669643fa158f2f64c1e6ab70837afbb9b0582e297f1da6e1c3c696fcd9`; the frozen harness SHA-256 is `094f43cb320f8e8e3196049269b20ac22e7e94fa9890b80f27f38ef49f7c82ea`; and the safe publication bundle SHA-256 is `51ac14bc3a6b4b570aaca6d4a41c91f53d9bf2887e66f0620c412f78566a3b44`.

The safe aggregates behind every number in Section 6 are `evidence/protected_v2/PUBLICATION_METRICS_V2.json`; the per-family contrast behind Figure 3 is `evidence/protected_v2/FAMILY_CONTRAST_V2.json`; and the result and custody attestation is `evidence/protected_v2/RESULT_ATTESTATION_V2.md`. Tables 1–3 and Figures 1–5 are regenerated from those two aggregates alone by `figures/generate_figures.py`, which never reads protected per-case labels. The mapping from every result-bearing sentence in this manuscript to the archived artifact that supports it is `evidence/CLAIM_LEDGER_V1.md`. The excluded exploratory arm is documented as `evidence/protected_v2/LIVE_ARM_STATUS.md`; its outputs are retained rather than deleted, and are not pooled with the protected campaign. The nearest-work dispositions that Section 2 discusses in prose are recorded per mechanism in `literature/NEAREST_WORK_AUDIT_2026-08-16.md`.

The safe bundle contains the post-run public attack slice and public per-case verdicts. Protected per-case gold and raw traces are retained separately in restricted custody and are not part of the public release.

10.3 Prospective naturalistic and comparator preflights

The registered naturalistic target is `protocol/P4_NATURALISTIC_IDENTIFIABILITY_SUCCESSOR_V1.json`. The metadata-only source snapshot is `development/naturalistic-panel-harness-2026-08-23/SOURCE_METADATA_SNAPSHOT_V1.json`, with canonical payload digest `7dd283acd81e63e84887d82ee7761e89e62a47d31afceca8628e456b76f73e01`; its exact P4 binding boundary is recorded in `development/naturalistic-panel-harness-2026-08-23/PROTOCOL_BINDING_STATUS_V1.json`. It contains repository, revision and licence metadata only, not raw cases, paired claim axes, gold, blinding records or outcomes. The provider- and modality-diverse successor frame is `development/provider-diverse-metadata-census-2026-08-23/SOURCE_CENSUS_V1.json`, with protocol `development/provider-diverse-metadata-census-2026-08-23/CENSUS_PROTOCOL_V1.json` and receipt `development/provider-diverse-metadata-census-2026-08-23/VERIFICATION_RECEIPT_V1.json`. It closes metadata structure only; exact historical-byte, content-rights, eligibility and case-panel authority remain unresolved.

The deeper attributed article-text candidate frame is archived as `development/p4-scientific-ascent-2026-08-23/source_binding/ARXIV_CC_BY_SOURCE_POOL_V1.jsonl`, with SHA-256 `47dd24657752731cdf45bab95852f3e18b50946af8a29b5acee95956ec81d895`. Its official OAI pages, record-level rights/identity binding, live diagnostic and structural receipt are stored beside it. The identity-complete natural-pair design is `protocol/P4_NATURALISTIC_IDENTIFIABILITY_SUCCESSOR_V2_AMENDMENT.json`; the executable outcome-free signal census and result are `development/p4-scientific-ascent-2026-08-23/screen_mechanism_feasibility.py` and `P4_NATURAL_PAIR_METADATA_FEASIBILITY_V1.json`. The provider/content expansion rule opened by the observed cell shortfalls is frozen at `protocol/P4_NATURAL_PAIR_SOURCE_EXPANSION_FREEZE_V1.json`. None of these objects contains an adjudicated case terminal or comparator outcome.

The first Zenodo route freeze and its preserved HTTP-400 transport receipt are `development/p4-scientific-ascent-2026-08-23/P4_ZENODO_RELATED_OBJECT_CENSUS_PROTOCOL_V1.json` and `development/p4-scientific-ascent-2026-08-23/P4_ZENODO_RELATED_OBJECT_CENSUS_RESULT_V1_RETAINED.json`. The distinct schema/pagination repair is `P4_ZENODO_RELATED_OBJECT_CENSUS_PROTOCOL_V2.json`; its disclosed pre-freeze identities, executable harvester, candidate rows, result and interpretation are stored beside it as `P4_ZENODO_V2_PREFREEZE_PROBE_DISCLOSURE.json`, `harvest_zenodo_related_object_census_v2.py`, `P4_ZENODO_RELATED_OBJECT_CANDIDATES_V2.jsonl`, `P4_ZENODO_RELATED_OBJECT_CENSUS_RESULT_V2.json` and `P4_ZENODO_RELATED_OBJECT_CENSUS_REPORT_V2.md`. Raw provider responses are byte-hashed but kept outside Git. The V2 result binds public source signals and their shortfall only; it contains no pair gold or model outcome.

The provider-disjoint DataCite M5 lineage is under `development/p4-datacite-m5-source-expansion-2026-08-23/`. The retained V1 protocol, empty candidate set, result, verification receipt and report are `P4_DATACITE_M5_CENSUS_PROTOCOL_V1.json`, `P4_DATACITE_M5_CANDIDATES_V1.jsonl`, `P4_DATACITE_M5_CENSUS_RESULT_V1.json`, `VERIFICATION_RECEIPT_V1.json` and `SCIENTIFIC_REPORT.md`. The distinct V2 exclusion disclosure, protocol, harvester, empty candidate set, result, verification receipt and report are `P4_DATACITE_M5_V2_PREFREEZE_DISCLOSURE.json`, `P4_DATACITE_M5_CENSUS_PROTOCOL_V2.json`, `harvest_datacite_m5_census_v2.py`, `P4_DATACITE_M5_CANDIDATES_V2.jsonl`, `P4_DATACITE_M5_CEN

SUS_RESULT_V2.json, VERIFICATION_RECEIPT_V2.json and SCIENTIFIC_REPORT_V2.md; their manifest is SHA256SUMS_V2. Raw public metadata response bodies are byte-hashed and kept outside Git. Neither identity retrieves descriptions, files, linked-publication text, labels, case outcomes or model outcomes.

The later multi-provider source-universe packet is development/p4-source-universe-successor-v3-2026-08-23/. It retains the frozen protocol, provider provenance and rights evidence, transport audit, lower-bound cell counts, bounded report, verification receipt, replay scripts and a local checksum manifest. To avoid committing reproducible public transport payloads, the 64,772,320-byte harvest bundle and 2,371-row candidate file remain outside Git; their SHA-256 digests and the full-source-manifest verification boundary are recorded in OMITTED_LARGE_ARTIFACTS.md. The packet contains no adjudicated natural pair, protected case, label or model outcome.

The distinct JOSS/GitHub M6 provider successor is archived under development/p4-m6-source-provider-successor-v4-2026-08-23/. Its pre-outcome protocol and freeze receipt are PROTOCOL_V4.json and PROTOCOL_FREEZE_RECEIPT_V4.json; the frozen Crossref page, 200-row metadata audit and provider-qualified projection are CROSSREF_PAGE_V4.json, CANDIDATES_V4.jsonl and STRICT_CANDIDATES_V4.json. Cell counts, rights/relation/transport evidence, the recursive negative-result ledger, bounded result and verification receipt are CELL_COUNTS_V4.json, RIGHTS_RELATION_TRANSPORT_AUDIT_V4.json, NEGATIVE_RESULT_LEDGER_V4.json, RESULT_V4.json and VERIFY_RECEIPT_V4.json. The local SHA256SUMS binds all 17 retained artifacts. The packet contains public metadata only and records 0 natural pairs, 0 exact paper-to-release relations, 0 author-lineage adjudications and no case, label or system outcome.

The no-extension exact-version bridge is archived under development/p4-m6-joss-exact-version-bridge-v5-2026-08-23/. Its protocol, frozen same-200 identities, archive and GitHub-resolution rows, 9/9 recovered V3 identities, cross-provider deduplication audit, exact bridge rows, cell counts, provider/lineage boundary, negative ledger, result and verification receipt are bound by the local SHA256SUMS. The packet records 39/80 exact public publication–archive-version–tag–commit bridges but 0 author-lineage adjudications and 0 natural pairs. It contains no case text, case label, system output, protected result, performance measurement or superiority result.

The same-identity content-addressed repair is archived under development/p4-m6-joss-bridge-repair-v6-2026-08-23/. Its frozen same-41 protocol, disclosed feasibility probe, row-level archive/commit identities, qualified SWHIDs, rights, negative ledger, result, harvest and verification receipts are bound by the local SHA256SUMS. The packet records 17/41 repairs and 56/80 final exact bridges, while retaining 24 unresolved identities, 0 author-lineage adjudications and 0 natural pairs. It contains no protected case, label, system output, performance measurement or superiority result.

The targeted same-identity V7 packet is archived under development/p4-unresolved-identity-v7-2026-08-23/. Its frozen 24-identity input, exact resolution rows, provider/rights harvest receipt, result, negative ledger and 25-check verification receipt are bound by the local SHA256SUMS. Fourteen identities are repaired, producing 70/80 exact bridges and leaving ten disclosed identities unresolved. The packet adds or replaces zero publication DOIs, performs zero lineage or natural-pair adjudications, and contains no protected case, label, model output, performance measurement or superiority result.

The exact ten-target V8 packet is archived under development/p4-unresolved-identity-v8-2026-08-23/. Its provider/SWH receipts, final adjudication and checksum manifest record three same-identity repairs, seven exact residuals and checksum-verified transport for all ten archives. No target source, comparator or scientific outcome is executed.

The V9–V11 exact-edge lineage packets are archived under development/p4-exact-edge-lineage-authority-v9-2026-08-23/, development/p4-exact-edge-lineage-authority-v10-2026-08-24/ and development/p4-exact-edge-lineage-authority-v11-2026-08-24/. V9 and the separately frozen V10B successor raise the bridge to 76/80; V11 retains the four exact provider-state discriminators and closes none. V11's result and directory-manifest SHA-256 digests are dee86c33a0347df5da2425ebc606c12020cfffdff6ac21cea6b23b570ad8fef4 and 72cb2fd2a3e0830fe113a69e8db2311416dab67b5cab59a1c732a20ea0048eed.

The index-91 build/provenance packet is archived under `development/p4-exact-edge-build-provenance-v12-2026-08-24/`. It retains the complete 106-class JAR projection, compiler failure, exact provider-mode receipts, SBOM, tool provenance, signing contract, negative-result ledger and 116-check validation receipt. Its result and `SHA256SUMS` SHA-256 digests are `ffb4a9b7a1482f18c87c157afc2fac8303dc59fa9511e086927f9eb5303b728a` and `c246007c2539e9bc8141bfbc286b444173160b0b01de86e5e567094c9f4e4b10`. No natural pair, protected label, comparator output, performance quantity or superiority result appears in these packets.

The live comparator identity/interface audit is `development/comparator-binding-harness-2026-08-23/COMPARATOR_BINDING_LEDGER_V1.json`, with bounded probes in `development/comparator-binding-harness-2026-08-23/PROBE_RECEIPTS_2026-08-23.json`. Those probes establish at most repository or entrypoint facts. No audited external system currently supplies the native three-terminal map required by the naturalistic endpoint, so the audit terminal remains `CANNOT_CHECK_FULL_COMPARATOR_FREEZE`.

10.4 Reproduction

The host-side scoring and separate-implementation reproduction paths are `host/evaluate_campaign_v2.py` and `host/independent_reproduce_v2.py`; the latter recomputes the headline counts from the safe bundle without sharing code with the scorer. This is not an external or institutionally independent replication. The binding manifest that ties subject, split, harness, and evaluator identities together is `reproducibility/BINDING_MANIFEST_V2.md`. Reproduction reconstructs the aggregates and the published figures and tables; it does not reconstruct the protected per-case gold, which is what the custody model deliberately withholds.

11 Conclusion

A scientific-verification result requires more than a score. The intended terminal must be identifiable from admissible scientific information, resistant to nuisance decoding, expressible by the compared interfaces, and capable of varying over the panel. The exact-risk and factorization results make these requirements testable: a target relation is implementable precisely when it is constant on the decision representation's fibres and its terminals are available; mixed fibres impose an irreducible lower bound.

The three P4 evidence layers instantiate different parts of that theory without rewriting one another. V2 eliminated false promotion across 360 hostile opportunities while preserving 60/60 clean promotions, but its H3 family was construction-saturated and remains NotSupported. V3 repaired the registered nuisance shortcuts and separated the exact CannotCheck terminal, but its comparison is principally one of interface expressiveness. P4-X then moved the comparator upward: after donor-complete provenance, verification, custody, version, epoch, and generic-authorization mechanisms were granted, the non-compensatory scientific-promotion relation scored 400/400 against 250/400 for the registered generic product and 50/400 for a compensatory product, while an information-equivalent typed product tied at 400/400. The tie is a positive boundary result: scientific promotion need not be centralized; it must be represented with target-sufficient state and the appropriate relation.

The resulting contribution is wider than a gate checklist but narrower than a deployed-system victory. It is a theory of when verification axes can support scientific claims, plus exact protected evidence that local verification and generic permission need not suffice for scientific promotion. The registered 768-cluster target is the next discriminator. A sixteen-root, provider- and modality-diverse two-wave metadata frame passes its structural checks, and a separate lawful arXiv frame fixes 1,536 attributed article candidates. The new source signal census prevents premature execution: six of eight mechanisms have at least one domain below the nominal pre-adjudication quota, so a frozen multi-provider expansion must run before a 32-cell panel can exist. Its first Zenodo identity retained a transport failure; a distinct repaired census then found only 29–44 publication-linked data records and 6–10 publication-linked software records per domain, below 48 in every frozen cell. The disjoint DataCite M5 successor repaired its first identity's client-field failure but found no frozen HTTPS content URL among 2,719 exact-rights metadata rows, so the M5 counts and deficits did not move. This not-reached relation count localizes the next work to repository-specific file manifests, repository-release/Software-Heritage and PMC OA routes rather than a weaker gate. A later

Dryad/Figshare/Harvard Dataverse/DataCite expansion bound 2,371 candidate rows, but incomplete frozen full-record transport, Dryad download and version-lineage failures, and zero repository-bound DataCite units leave its strict per-cell counts as lower bounds rather than confirmed deficits. The eight-cell source gate therefore remains unevaluable. A separately frozen JOSS/GitHub route transported 200 publications and identified 80 provider-qualified DOI/repository concepts. Its same-200 V5 bridge recovered 9/9 V3 M6 identities and established 39 exact public publication–archive-version–tag–immutable-commit relations. A distinct V6 same-identity repair used content-addressed archive/commit evidence to repair 17/41 failures without adding or replacing a publication, yielding 56/80 exact bridges. A targeted V7 exact-identity successor repairs another 14/24 and raises the bridge to 70/80; the remaining ten are partitioned across exact version, rights, archive–commit and tag–commit gates. These are source candidates, not eligible natural pairs: author-lineage adjudication, same-claim preservation and natural-pair identity remain unresolved. V7 creates no source-disjoint replication unit and cannot convert the earlier cell diagnostics into natural cases or global deficits. Author-lineage adjudications and natural pairs remain 0. The terminal `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK` remains unchanged. Eligibility, linked-object rights, external custody and exact executions of the four named comparator roles remain unresolved.

A ten-target V8 provider-native pass repairs three more same identities and reaches 73/80 exact bridges. V9 exact archive–revision comparison repairs two of the remaining seven, preserving the historical 75/80 stage. After its original target-specific test fails, a separately frozen outcome-informed V10B exact-content successor repairs one additional edge: the checksum-bound publication archive and a GitHub/Software-Heritage-authenticated immutable revision have identical 165-path manifests and MIT rights. The bridge is now 76/80, with four exact authoritative edges remaining. These passes create no lineage adjudication, eligible natural pair, source-disjoint replication, external custody, comparator execution or naturalistic outcome. V11 closes none of the remaining four and shows that further progress requires provider-state changes rather than another identical request. V12 supplies a sharper positive for index 91: all 106 archive-only Java classes are reproduced byte-for-byte by projecting the tracked JAR. It does not close the edge, because fresh compilation was not executable, signed build provenance is absent and two archive executable modes contradict the signed revision tree. The exact bridge consequently remains 76/80. Naturalistic, verifier-provider, source-disjoint, externally custodied and universal scientific-authorization superiority therefore remain open research targets, not promoted claims.

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A Freeze and custody identifiers

The final hidden set was created only after the public preflight and harness merge were signed and green. The independent host run is `31975937488`; the protected attack-set SHA-256 is `6b8b6243040163dfa582dbeb8896e2c68d6b263e3227b64e7bab199d8599897c`; and the candidate-manifest SHA-256 is `c5df341a6bf63f83d2af17492c498d4d36a944ad6616097a630d8390b7bda331`. Execution-freeze merge `99bcacc82224089c34019ad82287754388dadbc5` was GitHub-verified before launch. The launch workflow independently re-queried that signature and exact-main CI and byte-compared frozen workflow, binding, and harness files against the signed freeze commit before handoff.

B Ablation interpretation

Five 8.3% ablations each expose a hostile family governed by the removed coordinate. Source/provenance collapse exposes two families (16.7%), evaluator-protection removal exposes three (25%), and soft confidence permits promotion in eleven of twelve hostile families (91.7%). This monotone pattern is descriptive evidence that the final zero-false-promotion result is not generated by one redundant gate alone.

Table 4: Frozen Zenodo V2 publication-typed source signals after licence, public-file and pre-freeze-exclusion gates. The threshold is 48 in every cell; none passes. Counts are discovery candidates, not adjudicated pairs.

Domain	M5 article–data	M6 article–software
Earth/environment	37	6
Life/biomedical	44	10
Physical/engineering	29	10
Scientific software	30	7

C Prospective supplement: the source-expansion and transport-bridge programme

Nothing in this section is result-bearing. It records an outcome-free census and the state of a transport bridge that has not closed: the bridge stands at 76/80 and the programme has adjudicated **zero** natural pairs. It is reported because an unrun arm and an absent section look identical to a reader, and only one of them is a disclosure.

It sits outside the results deliberately. The counts below are lower bounds on observable candidates, not adjudicated pairs, and no claim in this paper rests on them. They describe what a future naturalistic comparison would have to draw from, and what still blocks it.

C.1 Outcome-free source-expansion census

The naturalistic successor was not executed. We instead ran a separate, outcome-free Zenodo feasibility census for its article–data (M5) and article–software (M6) source routes. The frozen V1 request failed before any census because Zenodo rejected its unauthenticated page size of 200 with HTTP 400; that transport terminal is retained. A distinct V2 successor disclosed and excluded every record identity seen in the transport/schema probes, used nine pages of 25 records for each of eight domain–mechanism queries, required an accepted open record licence and public file-link evidence, and counted only accepted relations whose target was publication-typed. All 72 responses passed the frozen schema and pagination checks, with no within-query duplicates.

V2 retained 173 query-specific candidates representing 98 unique Zenodo records. The publication-linked counts for the four M5 data queries were 37, 44, 29 and 30; the corresponding M6 software counts were 6, 10, 10 and 7. Thus none of the eight cells reached the unchanged pre-adjudication signal gate of 48. The contrast diagnoses different source bottlenecks: the M5 route is near but below quota, whereas M6 has many accepted typed relations but few publication-typed targets. The terminal is a source-cell shortfall requiring a disjoint provider route, not a natural-pair, scientific-performance or model outcome.

We then froze a provider-disjoint DataCite M5 successor without changing the four domains, 48-signal threshold, exact CC BY 4.0/CC0 declaration, HTTPS-content-URL requirement, publication-typed relation, or rights boundary. The first DataCite identity returned 4,000 valid metadata rows but omitted the client relationship required for provider disjointness; all rows therefore failed closed before rights or relation screening. Its terminal `P4_DATACITE_M5_CENSUS_V1_CANNOT_CHECK_TRANSPORT_OR_SCHEMA` is retained.

A distinct V2 mechanics repair bound and excluded all 3,479 unique DOI identities disclosed by V1, requested the client relationship, retained Zenodo source-family exclusions, and corrected only a literal documentation assertion. All four response schemas, pages and evidence assertions passed. Of 4,000 raw rows, 968 were excluded by the frozen V1 disclosure, leaving 3,032 provider-disjoint rows; 2,719 carried an exact CC BY 4.0 or CC0 declaration. None exposed an HTTPS `contentUrl`, so no row reached publication-relation adjudication and no candidate was admitted. The combined M5 counts therefore remain 37, 44, 29 and 30, while the separately preserved M6 counts remain 6, 10, 10 and 7. The terminal is `P4_NATURAL_PAIR_SOURCE_CELL_SHORTFALL__EXPAND_DISJOINT_SOURCE_UNIVERSE`. The zero relation count is a sequential not-reached count, not evidence that DataCite records globally lack publication relations. V2 repairs transport mechanics only; it is not a positive source, case or model result.

Table 5: Strict deduplicated public-source candidates observed before transport completion. Counts are lower bounds, not adjudicated natural pairs or confirmed quota failures.

Domain	M5 article–data	M6 article–software
Earth/environment	6	2
Life/biomedical	10	0
Physical/engineering	4	1
Scientific software	3	6

A separately frozen multi-provider expansion then audited Dryad, Figshare, Harvard Dataverse and DataCite for both M5 and M6. It retained 1,263 successful raw API responses, 26 provider-provenance captures and 2,371 candidate rows, while counting at most one candidate per concept DOI and linked publication DOI. Mirrors and metadata aggregators were not treated as independent providers. Admission required an exact accepted item licence, a structured publication relation and an unrestricted direct-download identity.

Provider transport was incomplete. Only 659 of 2,463 frozen Figshare full records and 512 of 1,540 Harvard Dataverse full records were captured; 1,804 and 1,028 identities, respectively, remain frozen rather than replaced. Dryad returned 1,200 metadata records, but anonymous download probes returned HTTP 401 and exact version/file lineage remained undetermined. DataCite supplied 2,958 discovery DOIs but zero counted units because the repository-specific record and file authority was not bound. The strict deduplicated counts in Table 5 are therefore lower-bound diagnostics, not confirmed cell deficits, and the eight-cell gate is not evaluable.

The source-feasibility terminal is `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK`. Natural-pair identity, author-lineage disjointness, same-claim preservation and scientific resolvability still require outcome-blind external adjudication. No case, model-performance or superiority result follows.

We next opened a separately frozen M6 provider route without altering any V3 identity. Its first Crossref page fixed 200 unique Journal of Open Source Software publications. All 200 JOSS pages transported; 191 exposed a labelled GitHub repository relation, 180 repositories exposed a latest release whose tag resolved to an immutable commit, 118 bound an accepted licence and licence-blob SHA at that exact tag, and 120 received one domain under the unchanged outcome-free token rule. The intersection contains 80 unique publication-DOI and repository concepts: 5 Earth/environment, 7 life/biomedical, 62 scientific software and 6 physical/engineering.

Those 80 are provider-qualified concepts for a later version-link and natural-pair audit, not eligible units. The JOSS page relates the paper to a repository, but this lane did not establish that the current GitHub release tag is the version evaluated by the paper. It therefore established 0 exact paper-to-release relations, 0 author-lineage adjudications and 0 natural pairs. The bounded V3 packet also omits its row identities, so cross-provider publication, repository and artifact deduplication is CannotCheck.

Even before those unresolved gates, the no-overlap union is only an optimistic diagnostic. Earth, life and physical each reach 7/48. Scientific software reaches 68/48, but JOSS/GitHub is one provider family and the disjoint Figshare side contains only 6/8 units. Thus neither surplus repositories nor releases, tags, assets, files, commits, versions, search hits or API responses can satisfy the frozen allocation gate. The V4 terminal is `P4_M6_JOSS_GITHUB_BOUNDED_TRANSPORT_COMPLETE__EXACT_PUBLICATION_RELEASE_VERSION_RELATION_AND_AUTHOR_LINEAGE_CANNOT_CHECK__M6_CELL_FRAME_NOT_READY`. The programme terminal remains exactly `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK`; no quota, source-frame, naturalistic or performance claim is promoted.

The no-extension V5 bridge then retained exactly those 200 JOSS publication DOIs and followed each labelled Software archive DOI through DataCite and source-native archive metadata. Of 200 archives, 103 exposed an explicit tag or commit identity for the same JOSS repository and 101 resolved to an immutable Git commit. After also requiring the original V4 provider-qualified identity, one distinct archive-version/concept relation, accepted exact archive and commit rights, unchanged repository/domain identity, source-native archive transport, and exact deduplication against 9/9 hash-recovered V3 M6 identities, 39/80 pass the publication–

archive-version–tag–commit bridge: 3 Earth, 4 life, 30 scientific-software and 2 physical concepts. Each is one unique publication-DOI/repository concept; files, tags, releases and commits do not multiply the scientific unit.

This closes an exact public transport relation for those 39, not natural-pair eligibility. The deduplicated V3-strict plus V5-exact candidate totals are 5/48, 4/48, 36/48 and 3/48; the Figshare replication sides are 2/8, 0/8, 6/8 and 1/8. All four gates fail in this exact transported frame; this does not convert the incomplete V3 provider census into a global population deficit. Author-lineage adjudications and natural pairs remain 0. Of the 80 V4-qualified concepts, 32 lack an archive-explicit same-repository tag/commit relation, 12 lack a distinct archive-version/concept relation and 3 lack accepted exact archive rights; overlap among failures leaves 41 incomplete. The V5 terminal is `P4_M6_JOSS_ARCHIVE_EXACT_VERSION_BRIDGE_PARTIAL_39_OF_80__M6_CELL_FRAME_AUTHOR_LINEAGE_AND_NATURAL_PAIR_CANNOT_CHECK`. It preserves the V4 and programme terminals and supplies no case, system outcome, naturalistic comparison or superiority evidence.

A prospectively separated V6 repair retained exactly the same 41 failed identities and added or replaced zero publication DOIs. Exact normalized source-archive/GitHub immutable-commit manifests repaired 12 identities; five additional identities were repaired by equality between a qualified source-native Software Heritage directory and the immutable Git commit root tree. Exact version/concept identity and accepted software rights at both the archive and commit remained mandatory. V6 repairs 17/41 and raises the exact bridge from 39/80 to 56/80: 4 Earth, 4 life, 45 scientific-software and 3 physical concepts. Files, versions, tags, commits, requests and manifests add zero scientific units.

The 24 unresolved identities comprise 12 exact-publication-version failures, 2 exact-archive-rights failures and 10 archive-to-commit content-identity failures under a mutually exclusive primary-cause assignment. Deduplicated V3-strict plus V6-final totals are 6/48, 4/48, 51/48 and 4/48; the unchanged Figshare replication sides are 2/8, 0/8, 6/8 and 1/8. Scientific software clears its total and JOSS-primary quotas but not its source-disjoint replication quota, while the other three domains remain below both gates. Thus every full cell fails, author-lineage adjudications and natural pairs remain 0, and the V6 terminal is `P4_M6_JOSS_SAME_IDENTITY_CONTENT_ADDRESSED_BRIDGE_REPAIR_PARTIAL_17_OF_41__FINAL_EXACT_56_OF_80__AUTHOR_LINEAGE_AND_NATURAL_PAIR_CANNOT_CHECK`. This is development transport evidence with a disclosed pre-freeze route probe, not confirmatory, global-provider, natural-pair, performance or superiority evidence. The programme terminal remains unchanged.

A targeted V7 successor then revisits only those 24 unresolved publication identities. It adds no publication DOI and does not count versions, tags, commits, files or requests as new units. Requiring an exact JOSS review relation, a unique publication-time archive child, a source-native tag resolved to a 40-hex commit, archive-authenticated content/origin identity, provider-bound checksums, and accepted rights at archive and commit repairs 14/24. The exact bridge rises from 56/80 to 70/80.

Ten identities remain unresolved under a mutually exclusive primary partition: three archive-to-commit authenticated-identity failures, two archive-software- rights failures, two exact archive-version DOI failures, and three exact tag-to-immutable-commit failures. Editorial assertions do not close these gates, and the packet performs zero replacement-publication, author-lineage or natural-pair adjudications. Its programme terminal is therefore `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK`. V7 is targeted development transport evidence, not a naturalistic case, model outcome or superiority result.

A still narrower V8 pass targets only those ten disclosed identities. Ten of ten provider archives are fetched once and checksum-verified; no publication is replaced and no file, version or request is counted as a scientific unit. Provider-native version, archive provenance and licence evidence repairs PSD, FAIRLinked and SEPARATE, raising the exact bridge from 70/80 to 73/80 in 36.011 measured seconds. Seven identities remain fail-closed, each with one exact missing authoritative edge. The pass adds no author-lineage adjudication, natural pair, replication unit, comparator execution or outcome, so the programme terminal remains `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK`.

A V9 same-identity successor subsequently repairs two of those seven edges by exact normalized checksum-bound archive–immutable-revision manifest equality, raising the bridge from 73/80 to 75/80. Of the five residual identities, a frozen V10 provider-authority pass leaves four unchanged and its target-specific test for the fifth fails as registered. A separately named, explicitly outcome-informed V10B protocol is then frozen before new archive download or comparison. For that fifth identity, an immutable later revision is bound by both GitHub and the repository's Software Heritage origin snapshot to the same Git tree as the publication archive; its checksum-bound Zenodo ZIP and GitHub codeload ZIP normalize to the same 165 paths, with zero missing, extra or differing files and byte-identical MIT licences. This exact-content witness raises the bridge to 76/80 without identifying the later revision as the deleted publication-version commit. Four authoritative edges remain. V9, V10 and V10B add no publication unit, lineage adjudication, natural pair, replication unit, comparator execution or outcome, so the programme terminal remains `P4_NATURAL_PAIR_SOURCE_TRANSPORT_CANNOT_CHECK`.

V11 revisits exactly the four remaining edges and closes none. The adverse result separates provider-state changes from repeated querying: index 36 needs a corrected checksum-bound archive whose root is the named publication commit; indices 133 and 185 need exact-file signed PyPI provenance to their full commits; and index 91 needs either exact payload equality or provider-native signed build provenance. Its cumulative bridge therefore remains 76/80.

V12 tests the index-91 causal split. The GitHub-authenticated immutable tree tracks `java/jPAI.jar` with SHA-256 `ff49482838e3a761913df327a48e819d4f9e552bfeab29655a82675c60c47162`. A frozen read-only projection yields exactly 106 class members, and their complete `java/bin/` multiset equals the 106 archive-only class files with zero missing, extra or differing bytes; all have Java class major version 61. This is a positive exact packaged-byte result, not a reproduced build. The pinned Java 17 OCI compiler did not execute because no Docker daemon was available and the disposable Colima fallback ended with the exact terminal `no_space_left_on_device`; no substitute compiler was used. Moreover, the validly signed revision tree records both `cpp/PAIpp.exe` and `run.sh` as mode 100644, while the archive records 0755, and no signed release or artifact attestation authorizes either change. V12 therefore closes zero edges and leaves the bridge at 76/80.